

Jeffrey William Shorthill

Salt Lake City, UT • 385-371-8083 • jeffrey.w.shorthill@gmail.com • github.com/jeffreywilliamportfolio

Summary

Independent AI interpretability and evaluation researcher who also ships production AI systems end to end. Seven publicly archived preprints (Zenodo DOIs) on mixture-of-experts routing, evaluation contamination, and safety behavior, built on preregistered, frozen-rule methodology: nulls reported, own claims retired, one headline confound self-refuted in public. Progressive Insurance's first AI Prompt Engineer. Solo-shipped a live dual-store consumer AI app and six deployed AI tools.

Selected Research (independent, public, reproducible)

- **In-Context Availability Reorganizes MoE Expert Routing** — cross-generation study (Qwen3.5 + 3.6-35B): teacher-forced designs isolate literal availability as a routing variable; attention-retrieval mechanism located and severed; frozen-prediction replication across a model generation; intervention nulls bound the claim. Direct implication for multi-turn *evaluation contamination*. DOI: 10.5281/zenodo.21892999.
- **When Routing Entropy Tracks Length, Not Complexity** — self-refutation of a position confound in routing-entropy analyses; the “complexity gradient” reverses under correct measurement (DeepSeek V3.1, Qwen3.5-397B). DOI: 10.5281/zenodo.20779602.
- **The Generation Half** — expert specialization is invisible in prefill and emerges during generation; rebuts a widely cited negative result; replicates at 122B. DOI: 10.5281/zenodo.20779604.
- **No Single Safety Gate** — harm-refusal routing is distributed; suppressing the lead expert reallocates with zero harmful completions. DOI: 10.5281/zenodo.20786891.
- **Dialect-Marked Response Audit** — matched-pair AAVE/AE safety-support equivalence; risk signal carried by register and action lexis. DOI: 10.5281/zenodo.20449546.
- **Register-readout artifact bundle** (e114-artifacts v1.3.1) — preregistered sweeps that *retired* the program's own earlier claims; all raw router archives, frozen analysis specs, and SHA-256 manifests public. DOI: 10.5281/zenodo.20709736.
- **Methodology track record** — preregistration and hash-frozen analysis rules; blinded/sham-controlled protocols; adversarial self-review; cross-runtime instrument validation; a fully blinded gravitational-wave echo search (sentinel-gated unblinding, positive-control gates) as transferable audit discipline.

Experience

Elevated Consultants LLC — Owner / Principal Feb 2024 – present
AI product consultancy; primary client: Gift Connect (nonprofit parenting program) Salt Lake City, UT

- Shipped the Gift Connect Parenting Suite (iOS + Android, live on both stores) and six deployed AI tools: realtime voice agent (OpenAI Realtime API/WebRTC), song and story generators, Twilio SMS assistant with CDC milestone tools, GA4 analytics dashboard, cohort portal.
- Sole engineer across mobile (React Native/Expo), web (Next.js), backends (Express, Flask/Celery), and infra (Vercel, Heroku, Firebase, Postgres), with consent-gated analytics.

Progressive Insurance — AI Prompt Engineer (2023; the company's first) 2018 – Dec 2023
Prior role: Customer Service Downtime Specialist (2018–2022)

- Proposed AI initiatives directly to executive leadership after GPT-4's release; a dedicated prompt-engineering role was created in response.
- Established internal prompting standards and taught applied AI sessions for data scientists, architects, and developers.

Webster Public Relations — Studio Manager, A/V Technical Specialist 2014 – 2018
Press-focused audio/video production Nashville, TN

Education & Skills

Full Sail University — A.S., Recording Engineering (2014) • **York County CC** — General Studies, Dean's List

Research: experiment design, preregistration, bootstrap inference, routing/activation capture (custom llama.cpp + HF hooks), Jacobian-lens readouts, SAE probes, intervention studies • **Engineering:** Python, PyTorch, TypeScript, React/React Native, Swift/SwiftUI, Next.js, CUDA-cloud ops (Vast.ai), OpenAI/Anthropic APIs, agent frameworks